

Giulia Bertagnolli

✉ giulia.bertagnolli@gmail.com | 🏠 gbertagnolli.github.io | 📧 gbertagnolli | 📺 giuliabertagnolli | 🎓 google scholar | orcid: 0000-0001-8637-0632

Experience

Researcher (RTD-a) SECS-S/01

FACULTY OF ECONOMICS AND MANAGEMENT, FREE UNIVERSITY OF BOZEN-BOLZANO

Development of statistical methods for the analysis of high dimensional and complex data.

Bozen, Italy

Dec 2024 - currently

Career break

MATERNITY LEAVE

June 2024 - Nov 2024

Researcher (RTD-a) SECS-S/01

DEPARTMENT OF MATHEMATICS (DIMA), UNIVERSITÀ DEGLI STUDI DI GENOVA

Development of statistical methodologies for (digital) health data. Funds by PNC DHEAL-COM.

Genua, Italy

Feb 2024 - Oct 2024

External Lecturer SECS-S/01

DEPARTMENT OF ECONOMICS, FREE UNIVERSITY OF BOZEN-BOLZANO

Undergraduate Statistics course.

Bolzano, Italy

Oct 2023 - Sept 2024

Post-doc

DEPARTMENT OF MATHEMATICS, UNIVERSITY OF TRENTO

Development of statistical methods for structured data, e.g. graphs and networks, distance and functional data.

Trento, Italy

Dec 2021 - Jan 2024

Education

PhD in Mathematics, summa cum laude

UNIVERSITY OF TRENTO & CoMuNe LAB, ICT, FONDAZIONE BRUNO KESSLER

- Thesis title: *Modelling the process-driven geometry of complex networks*
- Supervisors: Dr. Manlio De Domenico, Prof. Claudio Agostinelli

Trento, Italy

2019 - Dec. 2021

MSc. in Mathematics for Life Sciences

UNIVERSITY OF TRENTO

- Thesis title: *Complex Networks and Statistical Data Depths*
- Supervisors: Prof. Claudio Agostinelli, Dr. Manlio De Domenico
- with a 10-months internship at the CoMuNe Lab by the Bruno Kessler Foundation, Trento.

Trento, Italy

2016 - 2018

1st level Master in Data Science

BOLOGNA BUSINESS SCHOOL, ALMA MATER STUDIORUM - UNIVERSITÀ DI BOLOGNA

- Scholarship (total funding) from the Boston Consulting Group awarded on merit.
- 6-months internship at the TIM JOL SKIL Lab by the Bruno Kessler Foundations.

Bologna, Italy

2015 - 2016

Bachelor in Mathematics

UNIVERSITY OF TRENTO

- Thesis Title: *Common Principal Component Analysis with Applications*
- Supervisor: Prof. Pierluigi Novi Inverardi

Trento, Italy

2010 - 2014

Publications

A Weighted Likelihood Approach Based on Statistical Data Depths

C. AGOSTINELLI, A. BASU, G. BERTAGNOLLI, A. K. KUCHIBHOTLA

arXiv: 2507.16998

under review (JMVA)

2026

diffudist: Scalable Diffusion Geometry for Graphs via Spectral Methods

G. BERTAGNOLLI

CRAN package: vignette

submitted (JSS)

2026

Estimation of a multivariate von Mises distribution for contaminated torus data

G. BERTAGNOLLI, L. GRECO, C. AGOSTINELLI

arXiv: 2412.02333

under review (AStA)

2026

Detection and characterization of incidental vibrations from *Drosophila suzukii* in infested fruits

L. FELLIN, G. BERTAGNOLLI ET AL.

DOI: 10.1007/s10340-023-01711-8

Journal of Pest Science

2024

Efficient network exploration by means of resetting self-avoiding random walkers

G. COLOMBANI, G. BERTAGNOLLI, O. ARTIME

DOI: 10.1088/2632-072X/acff33

Journal of Physics: Complexity

2023

Dismantling the information flow in complex interconnected systems

A. GHAVASIEH, G. BERTAGNOLLI, M. DE DOMENICO

DOI: 10.1103/PhysRevResearch.5.013084

Physical Review Research

2023

Functional rich clubs emerging from the diffusion geometry of complex networks

G. BERTAGNOLLI, M. DE DOMENICO

DOI: 10.1103/PhysRevResearch.4.033185

Physical Review Research

2022

Unraveling the hidden organisation of urban systems and their mobility flows

R. GALLOTTI, G. BERTAGNOLLI, M. DE DOMENICO

DOI: 10.1140/epjds/s13688-020-00258-3

EPJ Data Science

2021

Diffusion geometry of multiplex and interdependent systems

G. BERTAGNOLLI, M. DE DOMENICO

DOI: 10.1103/PhysRevE.103.042301

Physical Review E

2021

Quantifying efficient information exchange in real network flows

G. BERTAGNOLLI, R. GALLOTTI, M. DE DOMENICO

DOI: 10.1038/s42005-021-00612-5, arXiv: 2003.11374

Communications Physics

2021

CovMulNet19: integrating proteins, diseases, drugs, and symptoms

N. VERSTRAETE, G. JURMAN, G. BERTAGNOLLI ET AL.

DOI: 10.1089/nsm.2020.0011

Network and Systems Medicine

2020

Network depth: identifying median and contours in complex networks

G. BERTAGNOLLI, C. AGOSTINELLI, M. DE DOMENICO

DOI: 10.1093/comnet/cnz041, arXiv: 1904.05060

Journal of Complex Networks

2019

Cities of a feather flock together: a study on the synchronization of communication between Italian cities

L. CANDEAGO, G. BERTAGNOLLI ET AL.

DOI: 10.1140/epjds/s13688-019-0198-4

EPJ Data Science

2019

Books

Multilayer network science: from cells to societies

O. ARTIME, B. BENIGNI, G. BERTAGNOLLI ET AL.

DOI: 10.1017/9781009085809, arXiv: 2401.04589

Cambridge University Press

2022

Recent Talks (last 5 years)

9th International Conference on Econometrics and Statistics – EcoSta 2026

INVITED

C. Agostinelli, A. Basu, G. Bertagnolli, A. K. Kuchibhotla. Weighted likelihood robust inference using statistical data depths.

Kyoto, Japan

August 8–10, 2026

International Conference of Robust Statistics – ICORS 2026

INVITED

Giulia Bertagnolli, L. Greco, C. Agostinelli. Robust Inference for Mixtures of Multivariate Wrapped Normal Distributions: The Weighted Likelihood Approach.

Istanbul, TR

July 20–24, 2026

Data Science, Statistics & Visualisation 2026 – DSSV2026

INVITED

G. Bertagnolli, Z. Huan, D. Ferrari. Sparse Unbiased Estimating Equations via Likelihood Score Alignment.

Trento, IT

June 29– July 1, 2026

5th ISI Regional Statistics Conference 2026

INVITED

Giulia Bertagnolli, L. Greco, C. Agostinelli. Estimation of a multivariate von Mises distribution for contaminated torus data.

Valletta, MLTA

June 3–5, 2026

y-SIS conference – StaTalk 2026

INVITED

Giulia Bertagnolli, G. Francisci, C. Agostinelli. Statistical data depths for network centrality and beyond.

Turin, Italy

May 21–22, 2026

19th International Conference on Computational and Methodological Statistics CFE-CMStatistics 2025

London, UK

INVITED

Dec 13–15, 2025

G. Bertagnolli, D. Ferrari. Approximate unbiased sparse estimating functions in high-dimensions

International Conference of Robust Statistics – ICORS 2025

Stresa, IT

CONTRIBUTED

May 19–23, 2025

Giulia Bertagnolli, L. Greco, C. Agostinelli. Estimation of a multivariate von Mises distribution for contaminated torus data.

Workshop of the Italian Chapter of the Complex Systems Society – CSS Italy

Online

INVITED

May 25, 2022

Giulia Bertagnolli, M. De Domenico. Functional rich-clubs in complex networks.

Conference on Complex Systems – CCS 2021

Lyon, FR

CONTRIBUTED

Oct 25–29, 2021

Giulia Bertagnolli, M. De Domenico. Diffusion geometry of multiplex and interdependent systems.

Joint Sunbelt and NetSci Conference – Networks 2021

Indiana University Network Institute,
US

CONTRIBUTED

Jul 5–10, 2021

Giulia Bertagnolli, M. De Domenico. Diffusion geometry of multiplex and interdependent systems.

International (Online) Conference on Complex Networks – CompleNet Live 2021

Exeter, UK

CONTRIBUTED

May 24–26, 2021

Giulia Bertagnolli, M. De Domenico. Diffusion geometry of multiplex and interdependent systems.

Organizer

Organized session at conference – Advancements in Statistics for Network and Metric Data

Berlin, DE

CFE-CMSTATISTICS 2026

December 12–14, 2026

The session aims to bring together recent methodological and applied contributions related to statistical inference, modeling, and computation for complex structured data, including network-valued, graph-based, and metric-space data.

Orgnizer of the Data Crunch at Lunch! Seminar Series

Free University of Bozen-Bolzano

SEMINARS

April 2025–currently

Co-orgnizer of the Research Seminar Series of the cluster DSMM

Free University of Bozen-Bolzano

SEMINARS

July 2025–currently

Teaching Activity

Data science applications for resource optimization, risk evaluation and sustainability

Free University of Bozen-Bolzano

TEACHING - MASTER IN DATA ANALYTICS FOR ECONOMICS AND MANAGEMENT

2025/2026 (24h)

Statistical Analysis of Network Data and Robust Statistical Methods. EduNext Format.

Statistical methods for business analysis

Free University of Bozen-Bolzano

TEACHING ASSISTANT - MASTER IN DATA ANALYTICS FOR ECONOMICS AND MANAGEMENT

2024/2025, 2025/2026 (18h)

Statistical inference and learning; model selection; regression, classification, clustering. Applications in R.

Introduction to Data Management and Data Analysis

Free University of Bozen-Bolzano

TEACHING - MASTER IN PUBLIC POLICY AND INNOVATIVE GOVERNANCE

2025/2026 (24h)

An introduction to Statistics and exploratory data analysis. Tabular, time-series, survey data.

Statistica [IT]

Free University of Bozen-Bolzano

TEACHING ASSISTANT - BACHELOR IN ECONOMICS AND MANAGEMENT

2025/2026 (18h)

Probability and Statistical Inference.

Data science applications for resource optimization, risk evaluation and sustainability

Free University of Bozen-Bolzano

TEACHING - MASTER IN DATA ANALYTICS FOR ECONOMICS AND MANAGEMENT

2024/2025 (18h)

Statistical Analysis of Network Data and Robust Statistical Methods.

Advanced statistics

Free University of Bozen-Bolzano

TEACHING ASSISTANT - MASTER IN DATA ANALYTICS FOR ECONOMICS AND MANAGEMENT

2024/2025 (18h)

Statistical inference, regression and classification; time-series analysis. Applications in R.

Statistical Methods for Biomedicine

TEACHING - BACHELOR "STATISTICA MATEMATICA E TRATTAMENTO INFORMATICO DEI DATI (SMID)"

Classification techniques and the logistic regression in the biomedical context.

Università degli Studi di Genova

April 2024 (20h)

Statistics

EXTERNAL LECTURER - BACHELOR IN ECONOMICS AND SOCIAL SCIENCES

Basic course in probability and statistics for the Bachelor in Economics and Social Sciences Details on my unibz webpage

Free University of Bozen-Bolzano

Oct 2023-Sept 2024 (48h)

The geometry of statistical models

TEACHING - MASTER IN MATHEMATICS

A short course (8h) to introduce some students of the master in Mathematics, to *Information geometry*. Details on my webpage

University of Trento

March 2023

Probability and Statistics

TEACHING ASSISTANT - BACHELOR IN COMPUTER SCIENCE

Teaching Assistant in the Probability and Statistics course for the Undergraduate degree in Computer Science

University of Trento

2018/2019, 2021/2022, 2022/2023

Graphical models and network science

INVITED LECTURER - MASTER IN MATHEMATICS

An introduction to network science and analysis, inside the course "Graphical models and network science" of prof. Veronica Vinciotti.

University of Trento

12h Dec. 2021

Fundamentals of network analysis: hands on

SEMINAR - MASTER IN DATA SCIENCE

Seminars for the course "Network Science: theory and lab".

University of Trento

(4h) 2021, (6h) 2020

Theses and Internships Supervision

MSc thesis co-supervision (MAT/06)

OF G. COLOMBANI

Colombani, Bertagnolli, Artime (2023)

University of Trento

2023

Internship supervision

OF A. M. CERASALE

Free University of Bozen-Bolzano

2024

Research Visits

Institute for Data Science Foundations, TUHH-with Prof. Nihat Ay

RESEARCH VISIT

Project on the information geometry of graphs, in collaboration with Prof. Giovanni Pistone, an expert in the field of Information Geometry.

Hamburg, Germany

April-May 2023

Nonparametric Information Geometry at DIMA, UNIGE

PHD LECTURE SERIES AND SEMINAR

PhD course by Prof. Giovanni Pistone for the research group of Prof. Eva Riccomagno. Details: giannidiorestino.it/HTMLs/2023-IG-Seminar.html

Genoa, Italy

Spring 2023

Summer & Winter Schools

Mathematical foundations of Quantum Machine Learning

DATASCIENCE.MATHS.UNITN.IT/EVENTS/QML2023

An introduction to the mathematical theory underlying Quantum Machine Learning, with three short courses held by professors G. De Palma, D. Trevisan, L. Banchi.

Trento, Italy

July, 10-14 2023

Machine Learning: From Data to Mathematical Understanding

C.I.M.E. COURSE 2003

The school focuses on the interplay between nonlinearity, nonsmoothness, and nonconvexity in high-dimensional probability and optimization and to acquire a deep understanding of relevant practical applications, such as image processing and classification. Lecturers: Schönlieb, Rigollet, Rosasco, Tropp, Wright.

Cetraro, Italy

July, 2-9 2023

Semiparametric Learning

DATASCIENCE.MATHS.UNITN.IT/EVENTS/SPL2021

An introduction to semiparametric methods in its general abstract form, and derive its detailed analytic forms in various applications including simple mean estimation, and more complex measurement error models, dimension reduction problems, missing data/causal inference problems. Lecturer: Yanyuan Ma.

Trento, Italy

July, 26-30 2021

Mediterranean School on Complex Networks

MEDITERRANEANSCHOOLCOMPLEX.NET/2019.HTML

Salina (Sicily), Italy

August, 31-September, 7 2019

The goal of this school is to provide a theoretical background to student (Master, PhD) and young researchers in the field, with particular attention to current trends in Network Science, and to promote philosophical and scientific exchange between all participants, lecturers and attendant.

Short courses

Trento, Italy

VARIOUS

2019-2023

I followed several short courses mostly in the fields of probability and statistics, among which “Asymptotics in statistics” by Alessandra Salvan, “Empirical processes” by Anand Vidyashankar, “An introduction to Malliavin calculus” by Luciano Tubaro, a reading-course on “Optimal transport”, “An introduction to Bayesian Non Parametrics” by Antonio Canale, “Algebraic statistics” by Kaie Kubjas, “Probabilistic Preference Learning with distance-based ranking models” by Valeria Vitelli.

Reviewer Activity

Documented peer review activity

WEB OF SCIENCE RESEARCHERID: GQB-0396-2022

- Information Geometry – Springer
- Physical Review Letters – American Physical Society
- Statistics Surveys – Institute of Mathematical Statistics
- Journal of the Royal Statistical Society: Series A (Statistics in Society) – Royal Statistical Society
- Physical Review E – American Physical Society
- Journal of Physics A: Mathematical and Theoretical – IOP Publishing (Selected as an Outstanding Reviewer for Journal of Physics A: Mathematical and Theoretical for 2019)
- Physical Biology – American Physical Society

Terza Missione

Le mille e una Scienza 2025

Bolzano, IT

PUBLIC OUTREACH

Nov 15 2025

Participation in outreach activities on Probability and Statistics, featuring “Il gioco della ProbabilOca”, with Greta Goracci.

Grafi, paradossi e sistemi complessi – Open Week 2024

Genua, Italy

UNIVERSITY (DIMA) OUTREACH

Feb 13, 2024

Notte dei Ricercatori e della Ricerca

Trento, Italy

PUBLIC OUTREACH

2019, 2021, 2023

Participation in outreach activities on complex networks and statistics. In 2023, development of the demo “Statistica dove non te l’aspetti”, showcasing statistical methods, including functional data analysis, for vibrational data.

Dipartimento di Matematica, University of Trento

Trento, Italy

UNIVERSITY OUTREACH

2019, 2022, 2023

Participation in orientation activities for high-school students, aimed at supporting informed choices of academic programs.